CSE 490/590 – Project 1: Checkpoint 2

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Date: October 18, 2026
Source code: https://github.com/ioleynik/CSE490

SourceCode zip file contains the main files

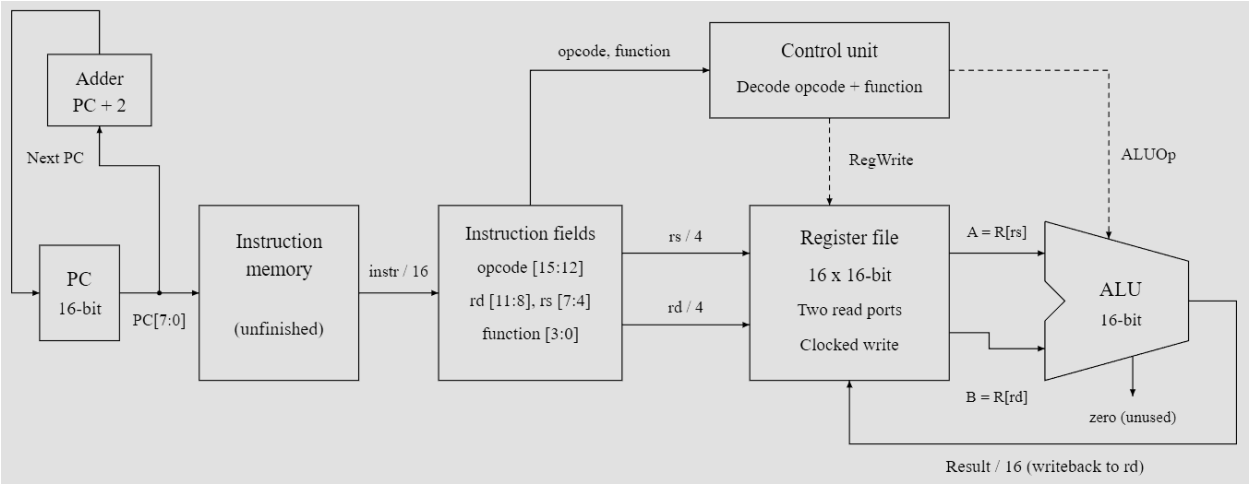
1. Current Progress

We have implemented and connected the main components for the R-type datapath of our 16-bit, single-cycle processor. Instruction memory and the remaining instruction paths are still being worked on. We worked on aligning the implementation with the required 16-register file, 4-bit register selectors, instruction-field layout, PC advancement by two bytes, and function-based R-type decoding.

Current files and their purposes:

- **pc.v:** Stores the current instruction address and supports synchronous reset.
- **reg.v:** Implements 1616-bit registers, 2 continuous read ports, and 1 clocked write port.
- **alu.v:** Implements ADD, SUB, SLL, AND, OR, XOR, and zero detection.
- **ctrl.v:** Decodes ADD, SUB, SLL, and AND using the opcode and function fields and generates ALU selection and reg-write enable.
- **cpu.v:** Connects the components, extracts instruction fields, and calculates PC + 2.
- **instructmem.v:** Declares instruction storage and its interface. Initialization and instruction output arent implemented yet

2. Current Datapath



3. Work Division focuses

- **Ilya:** ALU and register-file development, instruction-memory development, and planned data-memory implementation.
- **Faiyaz:** Program counter, control unit, top-level integration, and verification of the implemented components.
- **Both members:** Overall review, Architecture review, interface decisions, documentation, and remaining work.

4. Meeting and Discussion Log

- Sept 30 - Met up and discussed project idea and created Github Repo - Both Faiyaz and Ilya
- Oct 2 - Started work on the Project report and Verilog - Both Faiyaz and Ilya
- Oct 4 - Finished up the Project Report, Repo setup, and basic Verilog code - Both Faiyaz and Ilya
- Oct 8 - component development and decided on module interfaces - Both Faiyaz and Ilya
- Oct 9 - review and correction of register widths, instruction fields, and PC advancement - Both Faiyaz and Ilya
- Oct 13 - R-type decoding, ALU operations, and top-level integration - Both Faiyaz and Ilya
- Oct 17 -check the components tested, test method, and observed results - Both Faiyaz and Ilya
- Oct 18 - checkpoint submission prep and assignment of remaining work - Both Faiyaz and Ilya

5. Remaining Work

- Complete instruction-memory initialization and instruction fetching.
- Implement data memory, sign extension, and multiplexers for ADDI, LW, and SW.
- Add BEQ, BNE, and JMP control and address-selection paths.
- verify complete instruction execution, finish FPGA implementation, and prepare the final report.